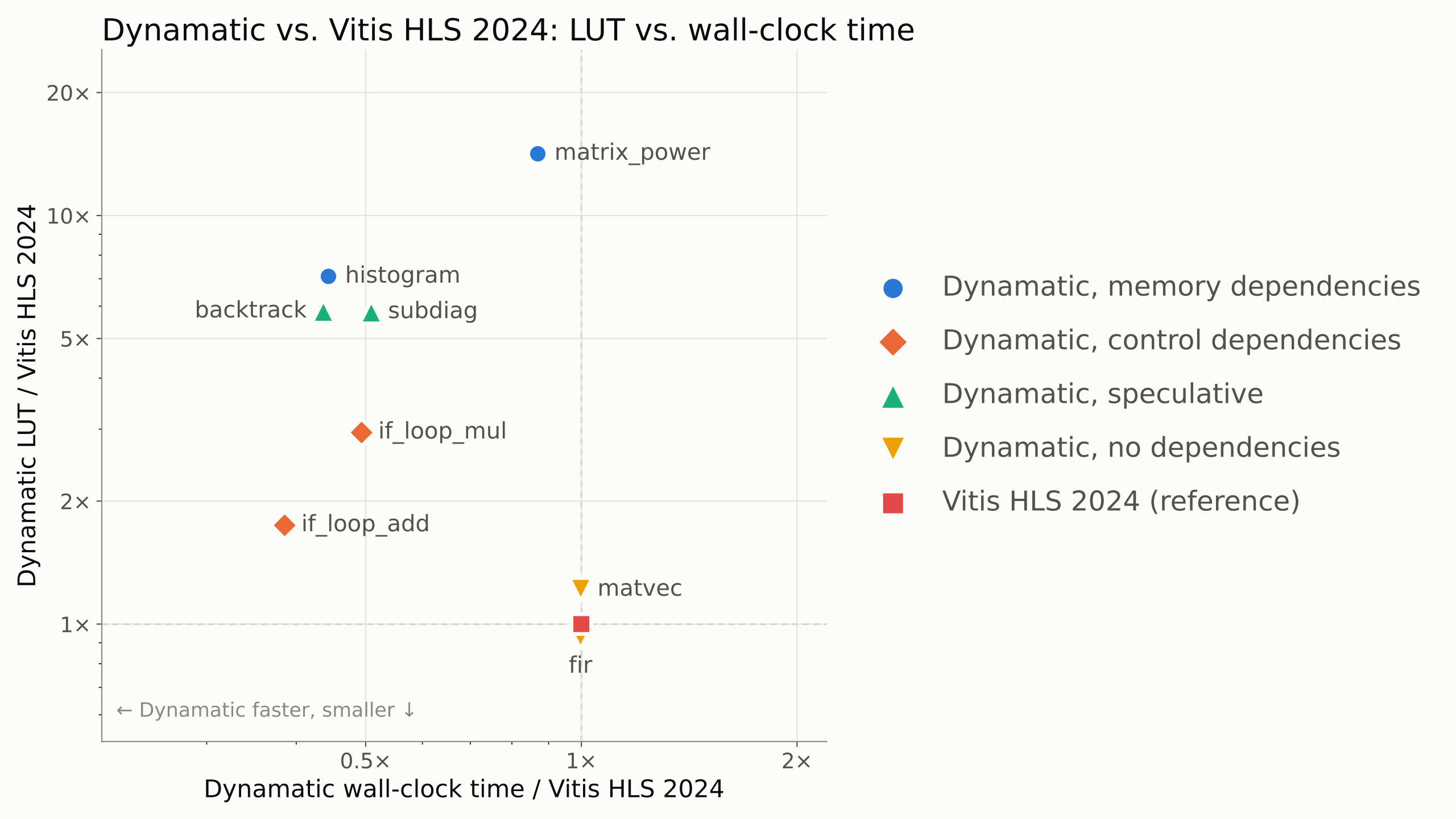
Dynamatic vs. Vitis HLS 2024: LUT vs. wall-clock time
20×
10×
5×
2×
1×
Dynamatic LUT / Vitis HLS 2024
matrix_power
histogram
backtrack
subdiag
if_loop_mul
if_loop_add
matvec
fir
← Dynamatic faster, smaller ↓
0.5×
1×
2×
Dynamatic wall-clock time / Vitis HLS 2024
Dynamatic, memory dependencies
Dynamatic, control dependencies
Dynamatic, speculative
Dynamatic, no dependencies
Vitis HLS 2024 (reference)